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Synchronized IoT Framework for Vibration-Based Structural Health Monitoring of Suspension Bridges

MUHAMMAD HASSAN [1] (Student Member, IEEE), AMR A. NASSR [2], AND MOHAMED ABDELRAHEEM [3] (Senior Member, IEEE)

[1]Electrical Engineering Department, Assiut University, Asyut 71515, Egypt
[2]Department of Civil Engineering, Ajman University, Ajman 346, UAE
[3]Department of Electrical Engineering and Computing, Rochester Institute of Technology—Dubai, Dubai 341055, UAE

CORRESPONDING AUTHOR: AMR A. NASSR (e-mail:a.nassr@ajman.ac.ae).

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ABSTRACT This article presents the design, implementation, and experimental validation of an Internet-of-Things (IoT)-based structural health monitoring (SHM) system for suspension bridges using synchronized vibration measurements. The proposed framework integrates distributed micro-electro-mechanical systems (MEMSs)-based sensing nodes, global positioning system (GPS)-assisted time synchronization, low-power wireless communication, and cloud-based modal analysis to support continuous monitoring of structural conditions. The main contribution of this work is the end-to-end integration of synchronized low-power IoT sensing, multitier wireless communication, and cloud-based vibration analysis within a unified SHM framework, together with experimental validation on a laboratory-scale suspension bridge. In the proposed system, each leaf node incorporates a high-accuracy digital accelerometer, ESP-NOW communication, and a duty-cycled sleep mode to reduce power consumption, while a central gateway aggregates the acquired data and transmits them to a remote server through a cellular network. Natural-frequency analysis, mode-shape evaluation, and the modal assurance criterion (MAC) are employed to assess changes in structural dynamic behavior. The system was evaluated using a laboratory-scale suspension bridge instrumented with five synchronized IoT sensing nodes and subjected to controlled shake-table excitations. Four hanger- and main-cable-damage scenarios were introduced to evaluate the overall operation of the proposed framework. The results demonstrate successful synchronized vibration acquisition, reliable wireless data transmission, and the ability to capture damage-sensitive modal changes, including natural-frequency shifts, mode-shape distortions, and MAC variations under different damage conditions. Communication experiments further show that ESP-NOW reduces energy consumption compared with transmission control protocol/internet protocol (TCP/IP) over Wi-Fi, extending node battery lifetime by approximately 40% through sleep-mode operation. Overall, the proposed SHM framework provides a cost-effective, energy-efficient, and synchronized IoT solution for vibration-based monitoring of suspension bridges under laboratory conditions. Future work will focus on field validation on real bridge structures and long-term operation under practical environmental conditions.

INDEX TERMS Cellular network, Internet of Things (IoT), structural health monitoring (SHM), synchronized sampling.

I. INTRODUCTION

Civil infrastructures, specifically bridges, contribute significantly to economic development and connectivity; however, these structures are subject to damage and degradation from environmental and operational effects. In cable suspension bridges, humidity-induced corrosion strongly influences the main structural elements by decreasing their load-carrying capacity [1]. In addition, fatigue damage resulting from cyclic traffic and wind loading may lead to significant deterioration of structural components, increasing the likelihood of failure

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and potential collapse [2]. Moreover, hangers[1] are crucial for maintaining structural stability, and their malfunction or failure can trigger severe vibrations and redistribute forces in a manner that threatens overall structural integrity.

Although classical methods such as visual inspection and routine nondestructive testing are useful, they are generally deficient in detecting minor internal damage. As a result, effective monitoring and maintenance procedures are vital. Structural health monitoring (SHM) systems provide real-time feedback on the status of a structure, allowing early intervention [3]. In addition, routine preventive maintenance mitigates risks associated with environmental and traffic-induced stresses [4]. Despite the advantages of proactive maintenance, its financial demands may discourage infrastructure investment, potentially leading to deferred maintenance and an increased probability of structural failure.

Recent advances in wireless communication, electronics, sensors, and the Internet of Things (IoT) have significantly enhanced the capabilities of SHM systems. By utilizing distributed sensor networks, IoT-based SHM enables continuous data acquisition, remote monitoring, and automated analysis of structural behavior. In addition, miniaturized sensors provide higher spatial and temporal resolution than traditional monitoring systems [5]. Wireless sensing technologies further reduce installation complexity and maintenance costs by eliminating extensive cabling requirements and minimizing manual inspection efforts. Despite these advantages, challenges remain regarding synchronization, energy consumption, data protection, and system scalability.

In SHM, a wide range of physical quantities can be measured, including strain, displacement, and acceleration. Recent reviews highlight that these sensing modalities are often combined to capture both local damage indicators and global structural response metrics in civil infrastructure [6]. Among these, vibration-response measurements acquired using accelerometers are widely adopted because they provide global information about structural stiffness and dynamic behavior without requiring direct access to the damaged region [7]. Recent studies have demonstrated that modern micro-electro-mechanical systems (MEMSs) accelerometers can provide sufficient accuracy for bridge and building monitoring while offering substantial advantages in cost, power consumption, and deployment flexibility for IoT-based SHM systems [8]. Combined with vibration-based damage detection (VBDD) techniques such as natural-frequency analysis, mode-shape evaluation, operational modal analysis (OMA), and the modal assurance criterion (MAC) [9], [10], [11], accelerometer-based monitoring provides an effective framework for assessing structural condition through vibration measurements.

Despite the maturity of VBDD algorithms, their practical deployment depends on the ability to acquire synchronized, high-quality measurements from distributed sensing nodes. Continuous vibration monitoring requires accurate time alignment across all sensing locations because timing mismatches can distort phase information and affect modal-parameter estimation. For example, a 1-ms synchronization error at 10-Hz introduces a 3.6° phase lag [12], [13]. To address this challenge, the proposed framework employs Global Positioning System (GPS) modules that provide pulse-per-second (PPS) synchronization signals for precise time alignment across all sensing nodes.

Reliable communication is another key requirement in IoT-enabled SHM systems. The communication architecture must support synchronized vibration acquisition while balancing throughput, energy consumption, scalability, and remote connectivity. These requirements become particularly important in battery-powered sensing systems, where communication overhead directly affects node lifetime and long-term deployment feasibility.

In this work, we present the design and development of an IoT-based synchronized SHM system for suspension bridges. The proposed approach integrates advanced VBDD techniques with state-of-the-art IoT sensing and communication technologies. This integrated solution is intended to support vibration-based structural condition assessment, proactive maintenance planning, and improved infrastructure management under the tested laboratory conditions.

The system adopts a distributed and time-synchronized vibration-monitoring methodology implemented through IoT technology. As illustrated in Fig. 1, the proposed framework consists of distributed sensing nodes, a central gateway, and a cloud-based monitoring server. The sensing nodes acquire synchronized vibration measurements from the bridge and transmit them to the gateway using a low-overhead wireless communication protocol. The gateway aggregates the collected data and forwards them through a cellular communication link to the cloud server, where modal analysis and structural-condition assessment are performed. Detailed hardware and communication specifications are presented in Section III.

The proposed method employs GPS-based synchronization to ensure accurate time alignment of acceleration measurements across all sensing nodes. In addition, it incorporates a multitier communication architecture combining short-range and long-range wireless links operating under a duty-cycled scheme to optimize the energy consumption of battery-powered IoT devices. This design enables reliable data transmission while preserving node lifetime through efficient power management.

We further validated the proposed framework through a series of laboratory experiments conducted on a scaled suspension bridge model. The experiments evaluate synchronized data acquisition, communication performance, power consumption, and the system's ability to capture damage-sensitive modal changes under representative structural damage scenarios.

Positioning and novelty: Unlike prior synchronized IoT-based SHM studies that focused either on generic

[1]Hangers are vertical or inclined cables or rods that connect the main cables to the bridge deck in a suspension bridge.

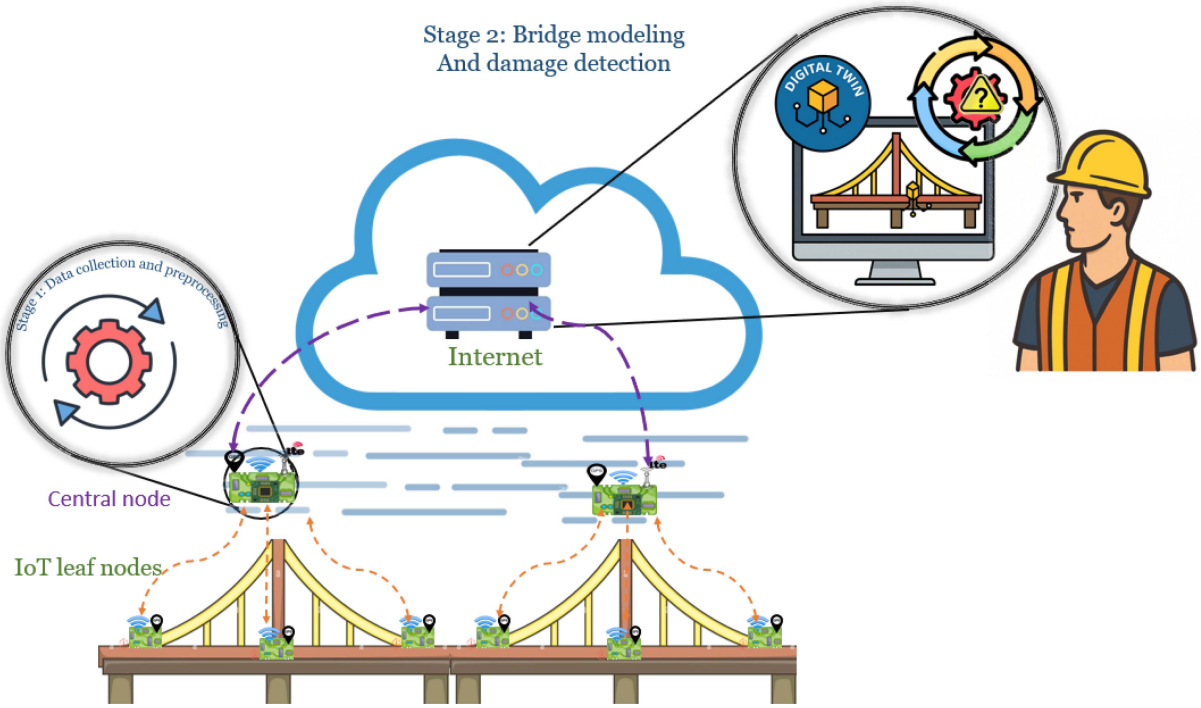


FIGURE 1. System architecture of the proposed IoT-based SHM framework. The IoT leaf nodes synchronously collect vibration signals from the bridge and transmit them to the central node using the ESP-NOW protocol. The central node aggregates and preprocesses the received measurements before forwarding them to the cloud server, where bridge modeling, modal analysis, and damage detection are performed to assess the structural condition.

synchronized sensing architectures or on cloud/fog data processing frameworks, this work presents an end-to-end SHM system specifically tailored to suspension bridges. Compared with our prior work, which established the general synchronized IoT-SHM concept and later explored fog/cloud computing integration, the present study combines GPS-PPS-based synchronization, low-overhead ESP-NOW local communication, long-term evolution (LTE)/TCP-IP backhaul, and cloud-based modal analysis into a unified pipeline, and validates it experimentally on a suspension bridge model under representative hanger- and cable-damage scenarios. Therefore, the novelty of this article lies primarily in the system-level integration, communication/power design, and bridge-specific experimental validation, rather than in proposing a new modal damage detection algorithm.

The rest of this article is organized as follows. Section II reviews the VBDD techniques and IoT technologies employed in the proposed framework. Section III presents the architecture of the developed IoT-based SHM system. Section IV describes the experimental setup and testing procedure. Section V presents the experimental results and performance evaluation. Finally, Section VI concludes this article.

II. DAMAGE DETECTION TECHNIQUES AND IOT SOLUTIONS FOR SHM

In this section, we provide an overview of the VBDD techniques employed in our system, and then, we review the IoT solutions commonly used for SHM applications.

A. DAMAGE DETECTION TECHNIQUES

The ability to identify damage relies on detecting variations in the structural dynamic behavior. Therefore, it is essential to monitor the changes in modal parameters such as natural frequencies, and mode shapes, as these parameters characterize the dynamic response. The extraction of these parameters requires capturing the forced or ambient vibration responses of the structure [14]. Compared to experimental (forced) modal analysis, which needs knowledge of the input excitation (typically provided by a mechanical shaker) [15], operational (ambient) modal analysis depends solely on the output response. In this case, the input is generated by environmental excitations such as wind, passing vehicles, or seismic activity [16].

Our system utilizes an automated OMA framework, where we adopt an automated scale-space OMA technique [17] to estimate the modal parameters. This technique involves computing the power spectral densities (PSDs) of the measured structural responses, followed by performing singular value decomposition (SVD) on the PSD [known as the frequency-domain decomposition (FDD) algorithm] to extract modal properties. Subsequently, a peak-picking algorithm [18] is applied to the eigenvalues and eigenvectors to determine the natural frequencies f (corresponding to the prominent peaks of the eigenvalues) and mode shapes Ψ [19]. Structural condition assessment and damage detection are achieved by observing the variation in modal parameters, since changes in mass or stiffness directly impact the mode shapes and cause shifts in the natural frequencies [20].

In the proposed system, we apply the following VBDD techniques.

1) *Natural frequency drop:* Damage within a structure causes changes in its physical and dynamic characteristics, notably resulting in a reduction of natural frequencies due to reduction in its stiffness. The natural frequency is highly sensitive to damage, where increased severity leads to a greater frequency drop. The variation in natural frequency is calculated using [21]

$$\Delta f = f_j^u - f_j^d \tag{1}$$

where f denotes the natural frequency, j refers to the jth mode, and u and d represent the undamaged and damaged conditions, respectively.

It should be noted that natural frequencies may decrease slightly even without structural damage due to changes in ambient or experimental conditions. Therefore, damage is considered likely when the frequency shift exceeds a threshold of 5% [22]. This method can only detect the presence of damage but not its precise location. However, it has the advantage of requiring fewer mode shapes for damage detection compared to the other techniques.

A quantitative estimate of damage severity can also be obtained from the natural frequency shift. Assuming that the modal mass remains approximately constant between the undamaged and damaged states, the equivalent stiffness ratio can be approximated as

$$\frac{k_d}{k_u} \approx \left(\frac{f_d}{f_u}\right)^2 \tag{2}$$

where k_u and k_d are the equivalent modal stiffnesses of the undamaged and damaged states, respectively, and f_u and f_d are the corresponding natural frequencies. Accordingly, an equivalent stiffness-loss index is defined as

$$\Delta k = 1 - \left(\frac{f_d}{f_u}\right)^2. \tag{3}$$

In this work, Δk is used as a complementary indicator of damage severity, while damage localization is still inferred from mode-shape changes and MAC analysis.

2) *Change in mode shape:* Mode shapes represent the deformation patterns of a structure under its natural frequencies. Structural damage alters the stiffness distribution, resulting in observable distortions in these patterns. Changes in mode shapes are particularly effective for localizing damage, as they reflect the structural response at specific locations. Visual inspection and quantitative comparison of the mode shapes—particularly in curvature—can highlight zones affected by damage. However, mode shape changes may be subtle for low-severity or symmetrically distributed damage and may require higher order modes for reliable detection.

3) *The MAC:* The MAC [10] is a statistical indicator used to assess the correlation between the mode shapes of the undamaged (reference) and test (possibly damaged) states. The MAC value is determined using the following equation:

$$\mathrm{MAC}_{j,k} = \frac{\left(\sum_{i=1}^{n} [\psi]_{i,j}^{u} * [\psi]_{i,k}^{d}\right)^2}{\sum_{i=1}^{n} \left([\psi]_{i,j}^{u}\right)^2 * \left([\psi]_{i,k}^{d}\right)^2} \tag{4}$$

where $\mathrm{MAC}_{j,k}$ represents the correlation between the $j$thand $k$thmode shapes. The indices j and k range from 1 to N and M, which are the number of modes in the healthy and damaged cases, respectively. $[\psi]_{i,k}^{d}$ and $[\psi]_{i,j}^{u}$ denote the components of the mode shape matrices (consisting of the first three modes) corresponding to the damaged and undamaged states. The subscripts u and d indicate undamaged and damaged conditions, and n is the number of sensor nodes.

The MAC value lies between 0 and 1, where a value close to 1 indicates high similarity between mode shapes, and a value near 0 implies no similarity. In practical scenarios, the MAC between mode shapes of the healthy (initial) state and those of the test state (assumed undamaged) typically yields values slightly less than 1 due to environmental uncertainties and numerical approximations [23]. An MAC value higher than 0.8 (commonly used as a rejection threshold) is generally considered acceptable and suggests a healthy condition of the structure. In contrast, MAC values below 0.8 indicate poor correlation, suggesting the presence of damage. However, this technique does not provide information about the damage location in relation to the sensor positions. Furthermore, a known limitation is that it requires higher order mode shapes for effective damage identification [24].

B. IOT TECHNOLOGY FOR SHM

In this subsection, we review different IoT technology implementations in SHM and discuss their capabilities and suitable use cases. This discussion justifies the need for a hybrid model that combines more than one IoT technology to achieve the targeted system goals. In recent IoT-enabled SHM systems, several communication technology classes have been adopted, each with different strengths and limitations. Wi-Fi-based solutions provide relatively high throughput and are suitable for transmitting vibration data, but they often involve higher communication overhead and power consumption. Within this class, lightweight protocols such as ESP-NOW are attractive for short-range, battery-powered sensing because they reduce protocol overhead. Cellular links such as 4 G/LTE, LTE-M, and related wide-area technologies provide long-range connectivity and direct cloud access, making them suitable for remote backhaul, although they are generally less attractive for deployment on every sensing node due to power and cost considerations. LoRa and LoRaWAN offer long-range, low-power communication, but their limited data rates make them more suitable for low-rate or event-driven SHM applications

TABLE 1. Comparison of Representative Recent IoT-Enabled SHM Systems Using Different Communication Technologies

Reference	Application / structure	Communication technology	System emphasis	Relevance to the present work
[25]	General civil infrastructure / low-cost vibration monitoring	HTTP over Wi-Fi / Internet	Uses smartphones and ESP32-based nodes for streaming vibration data to a Python platform with real-time feature extraction and unsupervised anomaly detection	Demonstrates a software-centric, low-cost IoT-SHM direction, but does not focus on synchronized multi-node modal SHM
[26]	Structural performance assessment with multi-sensor wireless monitoring	Wi-Fi via access point + GPS synchronization	High-throughput modular wireless monitoring system (MANTiS-32) for real-time structural assessment using multiple hubs and synchronized sensing	Relevant as a recent Wi-Fi-based synchronized SHM platform, but mainly oriented to high-throughput local monitoring
[27]	Laboratory-scale civil engineering structures	Wi-Fi (NodeMCU ESP8266)	Low-cost IoT vibration monitoring for real-time damage prediction and localization using DIY wireless sensor nodes	Relevant as a low-cost Wi-Fi SHM baseline, but limited to lab-scale validation and without hierarchical communication layers
[28]	Railway bridge SHM with digital twin	Local Wi-Fi + 4G gateway + MQTT	Hybrid edge–cloud architecture for high-frequency bridge vibration monitoring, anomaly detection, and digital-twin integration	Strong benchmark for hybrid communication architecture and for discussing the need to balance data rate, edge processing, and cloud analytics
[29]	Bridge SHM	NB-IoT	Low-power edge-computing nodes with embedded Random Decrement Technique to reduce transmitted data volume and extend autonomy	Relevant as a cellular LPWAN benchmark emphasizing energy efficiency and edge reduction rather than raw continuous streaming
[30]	Long-term bridge bearing monitoring	LTE Cat M1	Event-triggered vision-based monitoring system transmitting compact payloads to the cloud over LTE Cat M1	Shows that cellular-only SHM is practical when the payload is small, but it is not intended for dense synchronized vibration streaming
[31]	Footbridge vibration monitoring	LoRa (synchronous peer-to-peer nodes)	Two synchronous LoRa sensor nodes for low-cost modal identification and periodic vibration tracking	Relevant as a recent LoRa-based synchronized SHM example, though currently limited in node count and data richness
[32]	Historical masonry buildings	LoRaWAN	Long-term wireless monitoring of environmental and structural parameters such as humidity, crack width, and crack displacement	Relevant as a LoRaWAN-based SHM example for long-range, low-rate multiparameter monitoring rather than high-rate vibration transmission

than for continuous synchronized vibration streaming. Table 1 summarizes representative recent IoT-enabled SHM studies using these communication technologies and highlights the main tradeoffs among throughput, communication range, energy efficiency, and processing distribution. The purpose of this comparison is not to claim that one technology is universally superior, but rather to show that the communication choice depends on the functional role of the sensing layer and the backhaul layer within the SHM architecture. Based on this comparison, the proposed framework is positioned as a hybrid end-to-end SHM architecture that combines synchronized sensing, low-power local communication, long-range backhaul, and cloud-based modal analysis. Accordingly, the contribution of the present work lies mainly in the integration of these components into a unified suspension-bridge monitoring framework, together with bridge-specific experimental validation, rather than in proposing a new standalone communication protocol or damage-detection algorithm.

III. PROPOSED IOT-BASED SHM SYSTEM

This section describes the architecture of the proposed system, including the network topology, design of different types of IoT nodes, communication protocols, and the monitoring server.

A. LEAF NODES

The leaf node, illustrated in Fig. 2, is responsible for acquiring acceleration signals from the structure via an accelerometer and transmitting them to the central node. Each leaf node comprises the following components:

1) *Processing/communication unit:* The NodeMCU board (ESP8266 version), serves two key functions. First, it gathers acceleration data from the accelerometer and formats them appropriately. Second, it transmits these data to the central node.

   In this work, although the NodeMCU platform supports both standard Wi-Fi communication and the ESP-NOW protocol, ESP-NOW was selected to establish the link between the leaf nodes and the central node. This decision is motivated by the protocol's technical advantages in low-power IoT deployments. ESP-NOW is a lightweight, connectionless communication framework built on the IEEE 802.11 physical layer, providing comparable communication range and

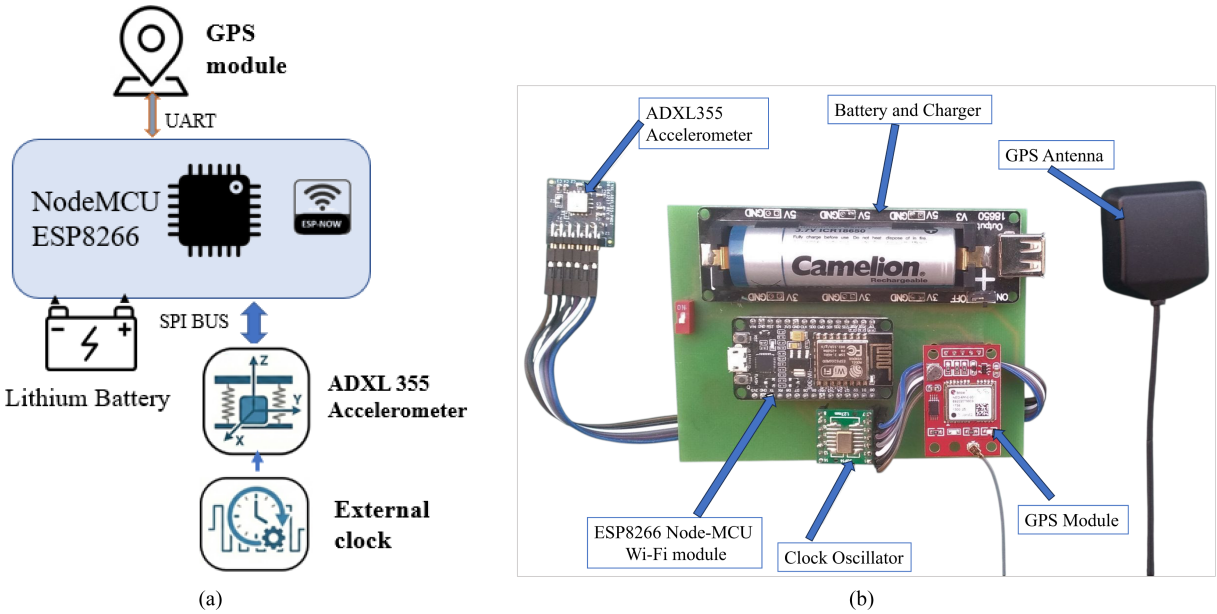


FIGURE 2. IoT leaf node. (a) Hardware architecture of the sensing node. (b) Physical implementation of the developed leaf node. (a) IoT system leaf node architecture. (b) IoT system leaf node.

sufficient data rate relative to Wi-Fi, while operating with substantially lower energy consumption. Unlike traditional Wi-Fi, ESP-NOW does not require association, authentication, or IP configuration, thereby reducing overhead and enabling ultra-low latency transmission. These characteristics make ESP-NOW particularly suitable for battery-powered sensing systems requiring fast, efficient, and reliable data exchange. The protocol's reduced complexity, multipeer capability, and robustness in congested 2.4-GHz environments further justify its integration into the proposed SHM architecture [33].

2) *Accelerometer sensor:* The ADXL355 is a digital, low-noise, low-drift, low-power three-axis MEMS accelerometer that integrates a 20-bit sigma-delta analog-to-digital converter (ADC) to digitize the filtered analog signals. It can record acceleration in three axes with output data rates ranging from 3.9 Hz to 4 kHz. An internal analog low-pass antialiasing filter limits the bandwidth to around 1.5 kHz. In addition, the device offers digital filtering options, including a low-pass decimation filter set at 31.25 Hz (with a sampling rate of 125 samples/sec) and a first-order high-pass filter with a 1.94-Hz cutoff. The accelerometer communicates with the NodeMCU using the SPI protocol. The SPI clock frequency is configured to 10 MHz, based on the NodeMCU's bus clock and baud rate settings.
3) *GPS module:* The GY-NEO6MV2 board featuring the u-blox 6 GPS module [34] is used for synchronizing the sampling time across all nodes. The PPS signal received from the GPS satellite system serves as a timing reference, enabling all leaf nodes to initiate data acquisition simultaneously.
4) *Battery:* Each node is powered by a 3.7-V, 2500-mAh lithium-ion battery, supported by a charger circuit. The battery can be recharged using a 10-W solar panel.
5) *Clock oscillator:* An external 1.024 MHz low-drift crystal oscillator [35] is used as the master clock for the ADXL355, replacing its internal high-frequency oscillator to enhance timing accuracy.

Table 2 summarizes the specification of IoT leaf-node components.

B. CENTRAL NODE

The central node receives acceleration signals from multiple leaf nodes, concatenates them, and forwards them to the remote monitoring server via the cellular network. It may also have its own accelerometer to capture local vibrations. The design of the central node, shown in Fig. 3, includes the following.

1) A processing unit (STM32F407 microcontroller) [36], a 32-bit ARM-based controller operating at 168 MHz.
2) An ESP8266 module responsible for short range communication between the leaf nodes and the central node.
3) The system incorporates a Quectel EC25 4 G LTE cellular module [37], interfaced with the microcontroller through a universal asynchronous receiver/transmitter (UART) communication link to provide reliable wide-area network connectivity over the internet. The EC25 enables long-range, IP-based, and high-availability communication between the central node and the cloud server by leveraging commercial cellular infrastructure.
4) An SD card used to temporarily store data in case of internet outage. Once connectivity is restored,

TABLE 2. Summary of IoT Leaf-Node Hardware and Communication Specifications

Component	Parameter	Specification
NodeMCU (ESP8266)	CPU architecture	Tensilica L106, 32-bit RISC
	Clock frequency	80 MHz (160 MHz overclock supported)
	SRAM	160 kB internal (instruction + data)
	Flash memory	Up to 16 MB (typically 4 MB)
ESP-NOW Communication	PHY layer	2.4 GHz IEEE 802.11 b/g/n Wi-Fi PHY
	Raw link bit rate	≈ 1 Mb/s
	Payload size	Up to 250 bytes per frame
	Peer capacity	Up to 20 peers (unencrypted + encrypted mix)
	Latency	Typically 1–3 ms
	LOS range	70–100 m (typical), 150 m+ optimized
ADXL355 Accelerometer	Sensor type	3-axis MEMS accelerometer
	Output resolution	20-bit sigma-delta ADC
	Noise density	$25\,\mu g/\sqrt{\text{Hz}}$ (typical)
	Analog LPF bandwidth	≈ 1.5 kHz
	Digital LPF	31.25 Hz (125 samples/s)
	High-pass filter	1.94-Hz cutoff
	Interface	SPI (up to 10 MHz)
GPS Module (u-blox NEO-6M)	Positioning update rate	Up to 5 Hz
	Time pulse (PPS) accuracy	$\pm 30\,ns$ RMS
	Cold start TTFF	27 s (typical)
	Sensitivity	-161 dB·m tracking
	Interface	UART (default: 9600 Bd)
External Clock Oscillator	Frequency	1.024 MHz
	Stability	Low-drift crystal oscillator
	Purpose	Master clock for ADXL355 (improved timing accuracy)
Power System	Battery type	3.7 V, 2500 mAh Li-ion cell
	Solar charging	10-W photovoltaic panel
	Charger circuit	Integrated Li-ion charge controller
	Estimated runtime	Application-dependent (low-power ESP-NOW optimized)

TABLE 3. Summary of Central-Node Hardware and Communication Specifications

Component	Parameter	Specification
STM32F407 MCU	Core architecture	ARM Cortex-M4, 32-bit RISC
	CPU frequency	168 MHz
	Floating point unit	Single-precision FPU
	Flash memory	1 MB
	SRAM	192 kB
Quectel EC25 4G LTE Module	LTE category	3GPP LTE Cat 4
	Downlink rate	Up to 150 Mb/s
	Uplink rate	Up to 50 Mb/s
	Supported bands	LTE-FDD: B1, B3, B7, B20, etc. (variant-dependent)
	Interfaces	UART, USB 2.0, PCM, SIM card interface

the buffered data are uploaded to the server. Table 3 summarizes the Central node main components specifications.

C. WIRELESS COMMUNICATION PROTOCOLS

Two wireless communication protocols are used in the system to implement the distributed sensing architecture. At the sensing layer, the leaf nodes perform synchronized vibration acquisition, local formatting, and packetized transmission of their measurements to the central node using the short-range ESP-NOW protocol. The central node operates as a gateway between the distributed sensor layer and the remote server: it manages data reception from multiple leaf nodes, aggregates the synchronized data streams, appends timing-quality information, and forwards the collected dataset to the cloud through a long-range Transmission Control Protocol/Internet Protocol (TCP/IP) link over the 4G cellular network. Therefore, the communication workflow follows the

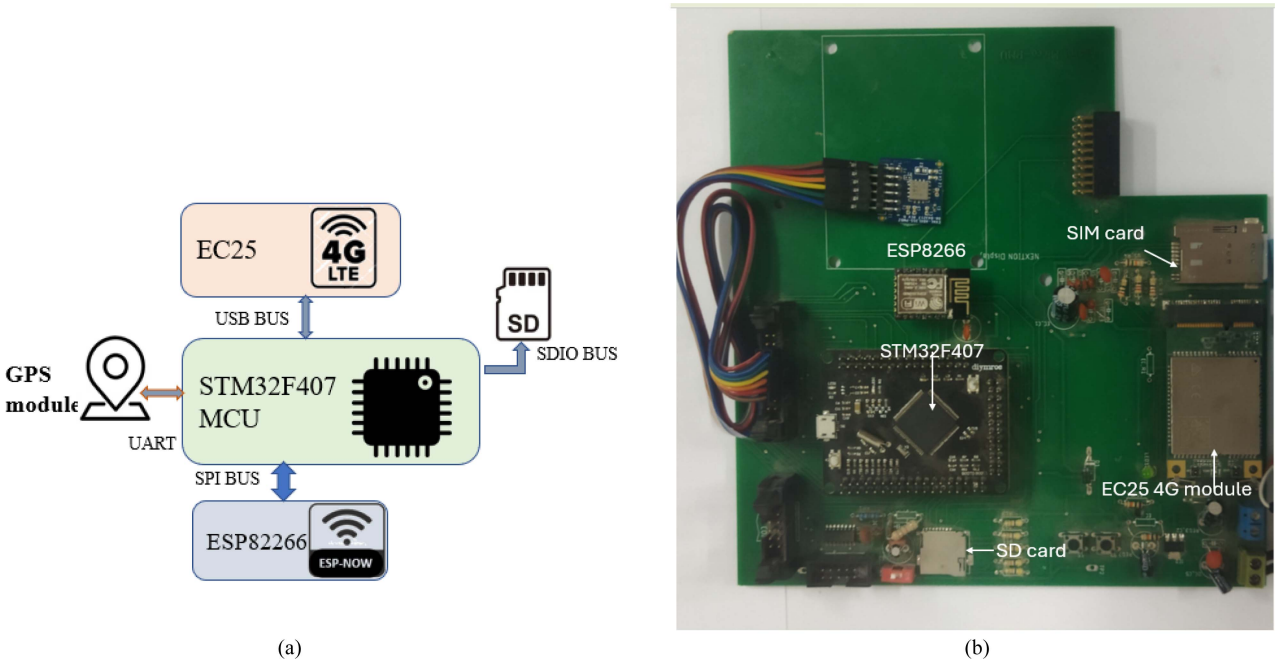


FIGURE 3. IoT central gateway. (a) Hardware architecture of the central node. (b) Physical implementation of the developed gateway node.

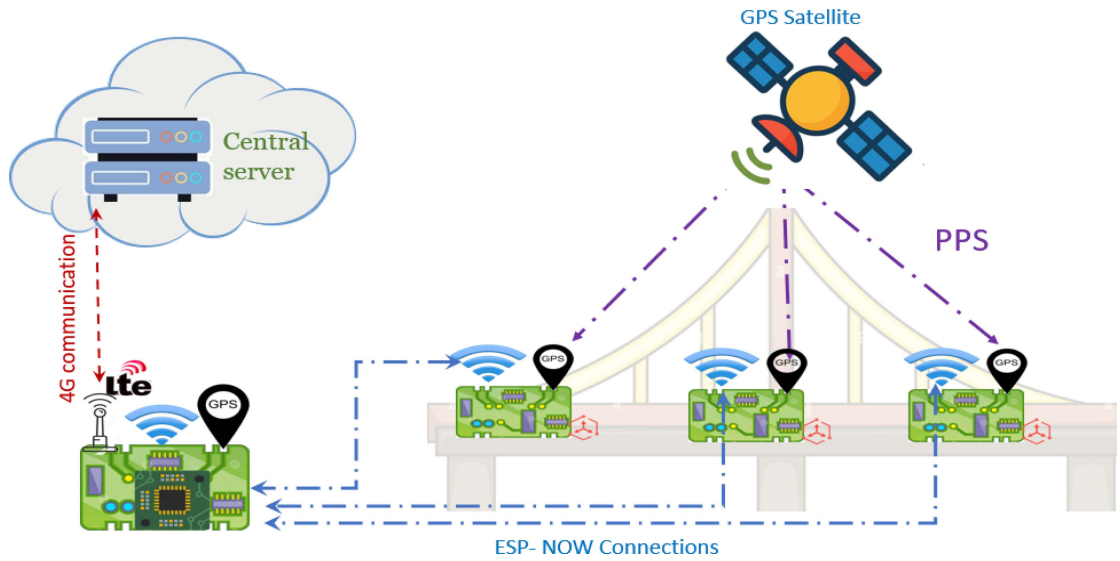


FIGURE 4. Communication architecture of the proposed SHM framework. GPS PPS signals provide synchronized timing references for all sensing nodes. Vibration measurements are locally acquired and transmitted to the central gateway through ESP-NOW, aggregated, and subsequently forwarded to the cloud server via the LTE cellular network for modal analysis and damage assessment.

sequence: synchronization, local acquisition, leaf-to-gateway transmission, gateway aggregation, and LTE upload to the server. Fig. 4 illustrates this communication topology.

1) *Short-range communication protocol:* Communication between the leaf nodes and the central node uses the ESP-NOW protocol [38], a peer-to-peer Wi-Fi-based protocol operating up to Layer 2 of the open systems interconnection (OSI) model. It does not require any handshake prior to data transfer and uses MAC addressing for sender and receiver. Each data frame can contain up to 250 bytes. Compared to TCP/IP over Wi-Fi, ESP-NOW is faster, more energy-efficient, and has less overhead because it bypasses the higher OSI layers [33]. In this system, communication between the leaf and central nodes is direct and point-to-point, with no need for IP routing

or TCP congestion and flow control mechanisms, at the same time, ESP-NOW supports packet integrity checks and retransmission, making it reliable for data transfer. In our implementation, the system uses a sampling rate of 125 samples/sec. Each sample is stored as a 4-byte float, requiring only two packets per second per leaf node. This batch approach minimizes packet overhead and reduces power consumption compared to transmitting individual samples.

However, the scalability of the local sensing layer is primarily constrained by the direct ESP-NOW communication range and the gateway peer capacity; therefore, for long-span bridges, the proposed architecture is more practically extended by deploying multiple central gateways, each serving a local cluster of leaf nodes and forwarding the aggregated data to the remote server.

2) *Long-Range Communication Protocol:* The central node uses a TCP/IP protocol over the cellular network to communicate with the remote server. The Quectel EC25 module transmits the collected data to the internet via the service provider's base station. Each transmission uses packets up to 1460 bytes, allowing all data collected in one second from all leaf nodes to be sent using just four packets.

The orchestration of ESP-NOW for short-range data collection and LTE TCP/IP for long-range transmission enables efficient, end-to-end communication between the sensing nodes and the remote server. By combining the low-overhead, fast local exchange of ESP-NOW with the reliable wide-area connectivity of the cellular network, the system ensures continuous and efficient data transfer from the field to the processing unit. Fig. 4 depicts the data transmission from the sensing node to the processing units.

D. MONITORING SERVER

The monitoring server runs on MATLAB [39] and uses a MySQL database for data storage. A TCP/IP listener on a predefined port establishes a persistent connection with the central node. Incoming packets are filtered, and acceleration data from each leaf node is extracted and stored. After receiving all data, the server performs several processing steps: computing the PSD, estimating modal parameters (natural frequencies and mode shapes), and applying the damage detection techniques introduced in Section II. The server also supports visualization of natural frequencies and mode shapes, and maintains historical data for long-term structural behavior analysis.

E. SYNCHRONIZED SENSING

In our design, each leaf node (with ADXL355 sensors sampling at 125 Hz) used the GPS PPS as a synchronization trigger. This ensured accurate alignment despite any delays introduced by microcontroller tasks. The PPS signal resynchronized the nodes every second. However, we observed that the internal clock of the ADXL355 exhibited up to 1% frequency variation, which could cause up to 20-ms drift per second. To address this, we replaced the internal oscillator with an external quartz-based phase-locked loop (PLL) oscillator, which reduced the timing error to just 50 μs. This significantly improved synchronization accuracy, enabling more reliable damage detection using vibration data. In practical outdoor deployment, environmental factors such as temperature-induced accelerometer offset, residual sampling-clock drift, and occasional GPS signal degradation may affect long-term performance; although the external quartz-based PLL oscillator substantially mitigates timing drift, further field validation is required to assess robustness under sustained environmental variability.

F. SLEEP MODE OPERATION

As the leaf node is battery-powered, extending its lifetime by reducing power consumption is essential. To achieve this, the NodeMCU enters a sleep mode[2] during idle periods. The node minimizes transmissions by grouping multiple samples into one packet, up to the Maximum Transmission Unit, sending only two packets per second. Fig. 5 shows a 1-s operation timeline for the NodeMCU. When the accelerometer completes a sample, it triggers an interrupt that wakes the NodeMCU. The processor stores the sample in a buffer, taking $T_N \approx 18\,\mu$s, then reenters sleep for $T_S \approx 7.982$ ms. Once the buffer is full, the processor stays awake to transmit data, which takes $T_{\text{TR}} \approx 0.643$ ms. After transmission, the node returns to sleep mode. This approach significantly reduces energy consumption and extends node battery life.

IV. EXPERIMENTAL SETUP

In this section, we describe the Bridge model and the series of experiments used to assess the system performance.

A. LABORATORY BRIDGE MODEL

The experimental evaluation is carried out on a laboratory-scale suspension bridge using Mola structure model [40]. The model includes two bridge towers, primary load-bearing (main) cables, and vertical suspension cables (hangers). Five accelerometer-based leaf nodes are installed, equally spaced, along the bridge deck to monitor vibrations.

The bridge model is placed on the Quanser Shake Table 1-40 [41], which provides controlled dynamic excitations such as sine waves, chirp signals, and an actual earthquake motion record as depicted in Fig. 6.

This setup is used to validate the ability of the proposed SHM system to detect and identify structural damage under various controlled loading conditions generated by the shake table.

The main experimental parameters used in this study are summarized in Table 4.

It should be noted that the experimental campaign was intended to validate the end-to-end operation of the proposed

[2]This system uses light sleep mode, where the CPU remains in a pending state while the system clock and communication module are disabled. Deep sleep mode, which fully powers down the CPU, is not suitable as it leads to loss of accelerometer data during wake-up. Thus, all references to "sleep mode" hereafter refer to light sleep.

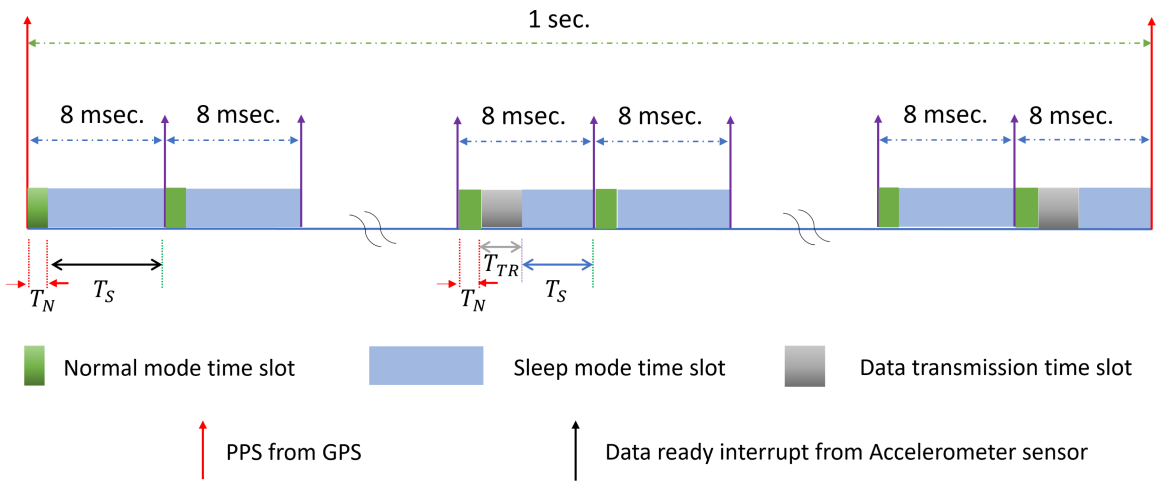


FIGURE 5. One-second timeline showing the different operating modes of the leaf node.

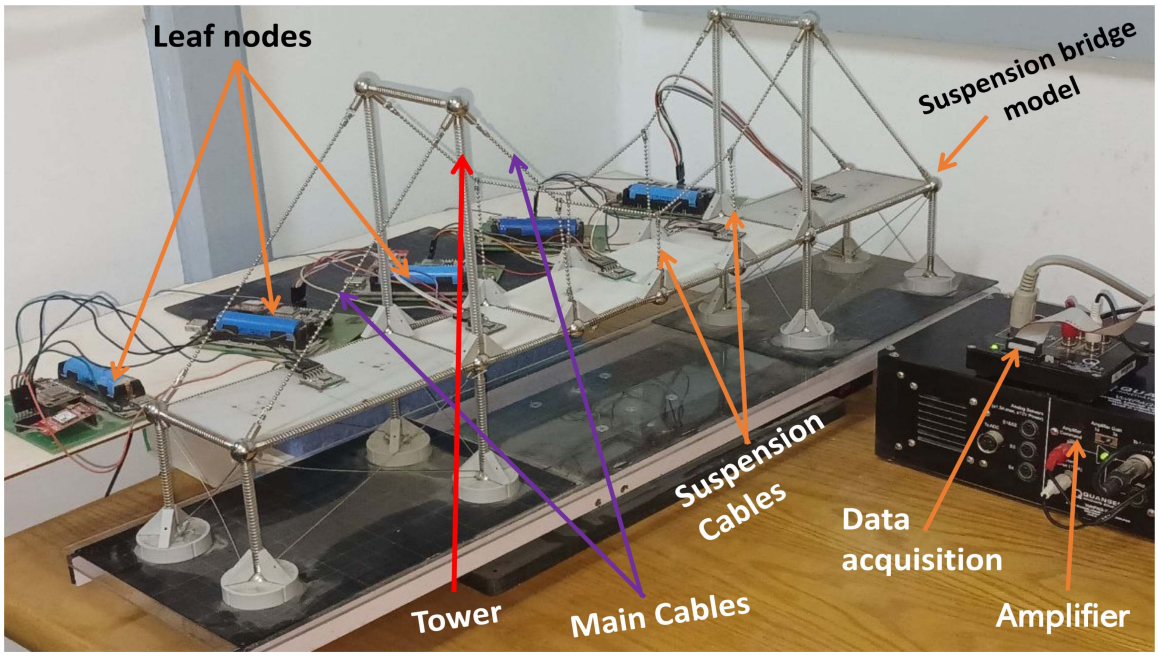


FIGURE 6. Experimental setup consisting of a laboratory-scale suspension bridge mounted on a shaking table and instrumented with five synchronized IoT sensing nodes.

synchronized IoT-SHM framework under controlled laboratory conditions. Although the structural states were evaluated more than once during system development, a formal statistical repeatability analysis was beyond the scope of this work. Therefore, the reported results are mainly used to demonstrate system functionality and the ability to capture damage-sensitive modal changes.

B. EXPERIMENTAL PROCEDURE:

The experimental program was designed to establish a reliable baseline of the bridge model in its undamaged condition and to evaluate its dynamic response under different damage scenarios. The procedure was carried out as follows.

1) Define the data acquisition parameters, including the number of samples and the sampling rate. Assign each leaf node a unique identifier.
2) Perform synchronized data collection to record the undamaged (healthy) state of the bridge.
3) Apply external excitations to the bridge model using the shaking table.
4) Collect vibration data synchronously from all leaf nodes using GPS-based PPS signals and external low-drift clock references.
5) Transmit the sampled data from each leaf node to the central node using the ESP-NOW short-range communication protocol.

TABLE 4. Summary of Main Experimental and Preprocessing Parameters

Parameter	Value / description
Number of sensing nodes	Five leaf nodes
Sensor placement	Equally spaced along the bridge deck
Sensor	ADXL355 three-axis MEMS accelerometer with 20-bit sigma-delta ADC
Sampling rate	125 samples/s
Window size	2048 sample
Window Type	Hann
Window overlap	50%
FFT length	2048
Analog antialiasing filter	Internal LPF, bandwidth $\approx$ 1.5 kHz
Digital low-pass filter	31.25 Hz
High-pass filter cutoff	1.94 Hz
Synchronization reference	GPS PPS
External clock oscillator	1.024-MHz low-drift crystal oscillator
Timing drift before improvement	Up to 20 ms/s
Timing error after improvement	About 50 μs

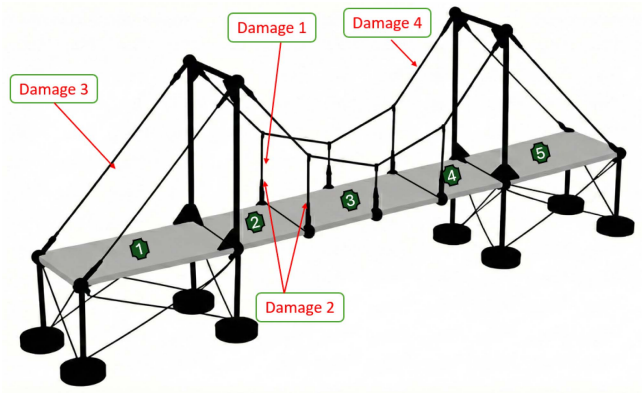


FIGURE 7. Locations of the sensing nodes and the four artificial damage scenarios introduced in the suspension bridge model.

TABLE 5. First Three Natural Frequencies of the Suspension Bridge Model

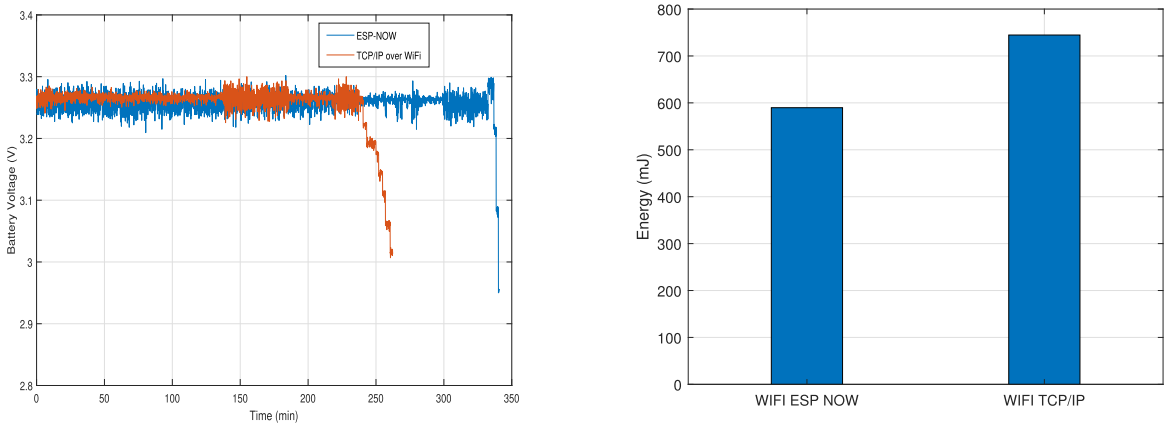
State \ Freq.	1	2	3
Undamaged	7.8125	10.659	17.807
Damage_1	5.9814	10.6811	17.761
Damage_2	4.0894	6.8969	10.681
Damage_3	5.7373	8.2397	10.437
Damage_4	4.2114	8.1176	10.620

6) Relay the aggregated data from the central node to the remote monitoring server through the cellular Internet.
7) Store the received data in a structured database and perform modal parameter extraction to establish a baseline of the bridge's healthy condition.
8) Introduce artificial damage by removing one of the main cables or suspension cables from the bridge model.
9) Repeat the data acquisition and processing steps to record the dynamic response of the damaged structure.
10) Apply the VBDD methods on the monitoring server to identify and locate the inflicted damage.

For validation purposes, four distinct damage scenarios were applied as follows.

1) *Damage 1:* One hanger between Sensors 2 and 3 was removed. This damage primarily reduces the local vertical stiffness of the deck-hanger subsystem and is therefore expected to produce a localized deformation pattern, with only a moderate influence on the global modal properties.
2) *Damage 2:* Two hangers between Sensors 2 and 3 were removed on opposite sides. This configuration introduces a more severe and asymmetric local stiffness loss, which is expected to cause stronger mode-shape distortion, greater perturbation of the natural frequencies, and possible torsional coupling effects.
3) *Damage 3:* The edge segment of the main cable between Nodes 1 and 2 was removed. Unlike hanger removal, this damage weakens the boundary-related stiffness near the tower–cable interface and is expected to affect the global dynamic response through stiffness redistribution, while still showing some localization near the damaged edge.
4) *Damage 4:* The main cable between the two towers was removed. Since this cable is one of the primary load-bearing elements of the suspension system, its removal produces the most pronounced global stiffness reduction and force redistribution along the span, and is therefore expected to induce the largest overall change in modal response.

The four damage scenarios were selected to represent a balanced range of cable-related damage mechanisms, spanning localized hanger-stiffness loss to more global main-cable degradation. This enables observation of how different physical damage mechanisms influence the bridge dynamic response in terms of natural frequencies, mode shapes, and modal correlation. In the results section, the relative severity of these scenarios is further quantified using the equivalent stiffness-loss index derived from the first-mode frequency shift, while the associated mode-shape and MAC variations are used to interpret the spatial characteristics of the damage. The node placement, numbering, and artificial damage locations are depicted in Fig. 7.



(a) (b)

FIGURE 8. Communication-energy evaluation. (a) Battery-voltage decay of the leaf node during continuous transmission using ESP-NOW and TCP/IP over Wi-Fi. (b) Energy consumed per packet transmission for both communication protocols.

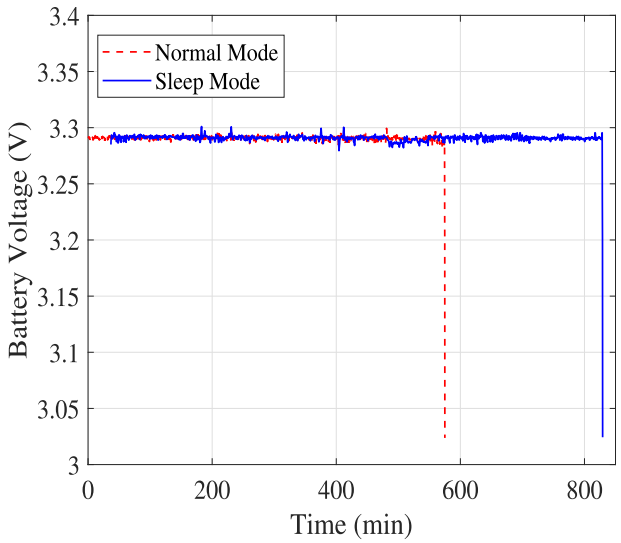


FIGURE 9. Comparison of leaf-node battery lifetime under normal operation and sleep-mode operation.

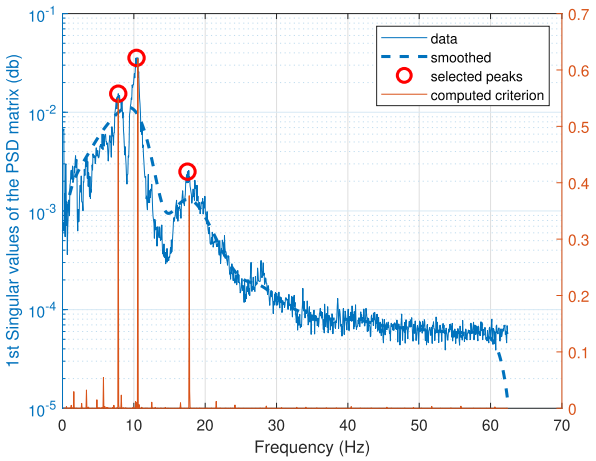


FIGURE 10. PSD of the undamaged bridge model showing the dominant frequencies identified using the scale-space peak-picking algorithm.

TABLE 6. Equivalent Stiffness-Loss Index Derived From the First-Mode Frequency Shift

State	f_1 (Hz)	$(f_d/f_u)^2$	Δk
Undamaged	7.8125	1.000	0.0%
Damage 1	5.9814	0.586	41.4%
Damage 2	4.0894	0.274	72.6%
Damage 3	5.7373	0.539	46.1%
Damage 4	4.2114	0.291	70.9%

V. PERFORMANCE EVALUATION

This section presents the performance evaluation of the proposed IoT-based SHM system using experimental results. In addition to assessing the operational characteristics of the IoT framework, including power consumption and transmission efficiency, the primary objective is to validate the system's effectiveness in detecting and identifying structural damage. This is achieved through vibration-based analysis employing mode shapes, natural frequencies, and the MAC technique. It should be noted that the present results are intended primarily to validate the system-level operation of the proposed synchronized IoT-SHM framework under controlled laboratory conditions. A comprehensive statistical repeatability analysis was beyond the scope of the current study and is considered an important direction for future work.

A. TRANSMISSION EFFICIENCY

This experiment evaluates the transmission efficiency of ESP-NOW compared to the conventional TCP/IP stack over Wi-Fi for short-range communication. Although the ESP8266 typically utilizes a full four-layer TCP/IP protocol stack, such complexity is unnecessary for the proposed system. Since there is no need for routing or maintaining persistent connection channels, relying on TCP/IP introduces additional transmission overhead and redundant handshaking processes. In our system, the leaf node uses ESP-NOW layer two

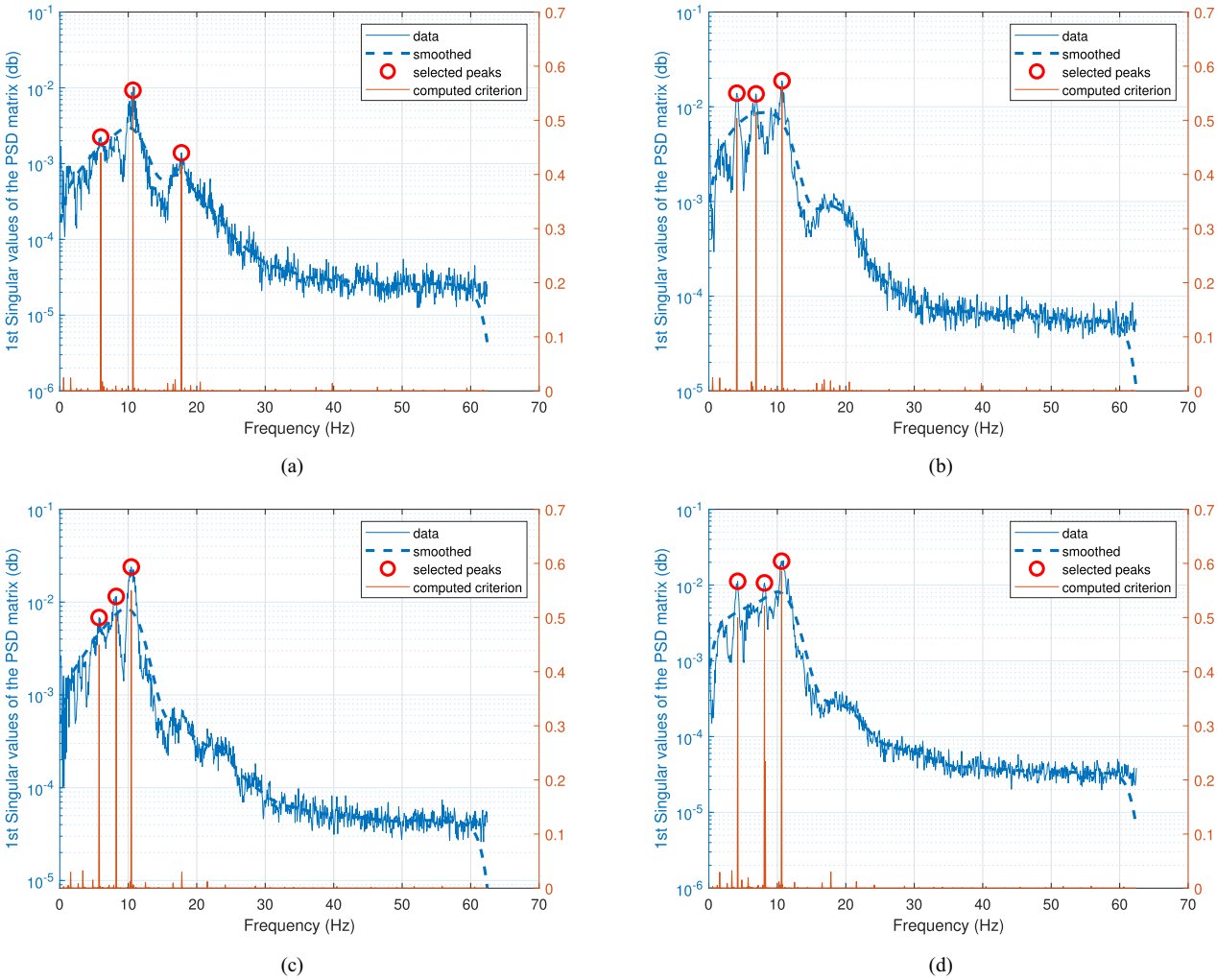


FIGURE 11. PSD of the suspension bridge under different damage scenarios. Dominant frequencies were identified using the scale-space peak-picking algorithm. (a) Damage scenario 1 (one suspension cable removed). (b) Damage scenario 2 (two suspension cables removed). (c) Damage scenario 3 (Edge main cable removed). (d) Damage scenario 4 (Main cable removed).

protocol to facilitate direct transmission between the leaf and central nodes.

To highlight the advantages of using ESP-NOW, a lightweight layer two communication protocol, we conducted an experiment that compared battery voltage decay over time when transmitting an identical number of packets using both protocols. In this setup, the leaf node periodically transmits dummy packets using each protocol (in two separate runs). Simultaneously, a secondary microcontroller connected to the battery's ADC continuously monitors and logs the voltage level.

As illustrated in Fig. 8(a), ESP-NOW maintains the battery's rated voltage for a significantly longer duration due to its streamlined communication process. In contrast, TCP/IP over Wi-Fi results in faster battery depletion. Once the battery voltage drops below 3.1 V, the leaf node shuts down, halting further transmission.

In addition, Fig. 8(b) presents the energy consumed during one second of continuous data transmission, clearly demonstrating that ESP-NOW is more energy-efficient than Wi-Fi-based TCP/IP communication.

Fig. 9 shows the battery lifetime when the leaf node processor (Node MCU) operates in a normal mode all the time and when the sleep mode is enabled. Implementing the sleep mode extended the node lifetime to around 40% compared to the normal mode operation.

B. NATURAL FREQUENCY ANALYSIS

The first step is to calculate the PSD of the acceleration and use the scale-space peak detection algorithm to delineate the value of the natural frequencies. The synchronized multisensor responses were analyzed using the standard FDD procedure. Specifically, the auto- and cross-PSDs were first estimated using Welch's method to form the PSD matrix at each frequency line, with the adopted window, segment length, overlap, and fast Fourier transform (FFT) length reported for reproducibility. SVD was then applied to this matrix, and the peaks of the first singular value were used

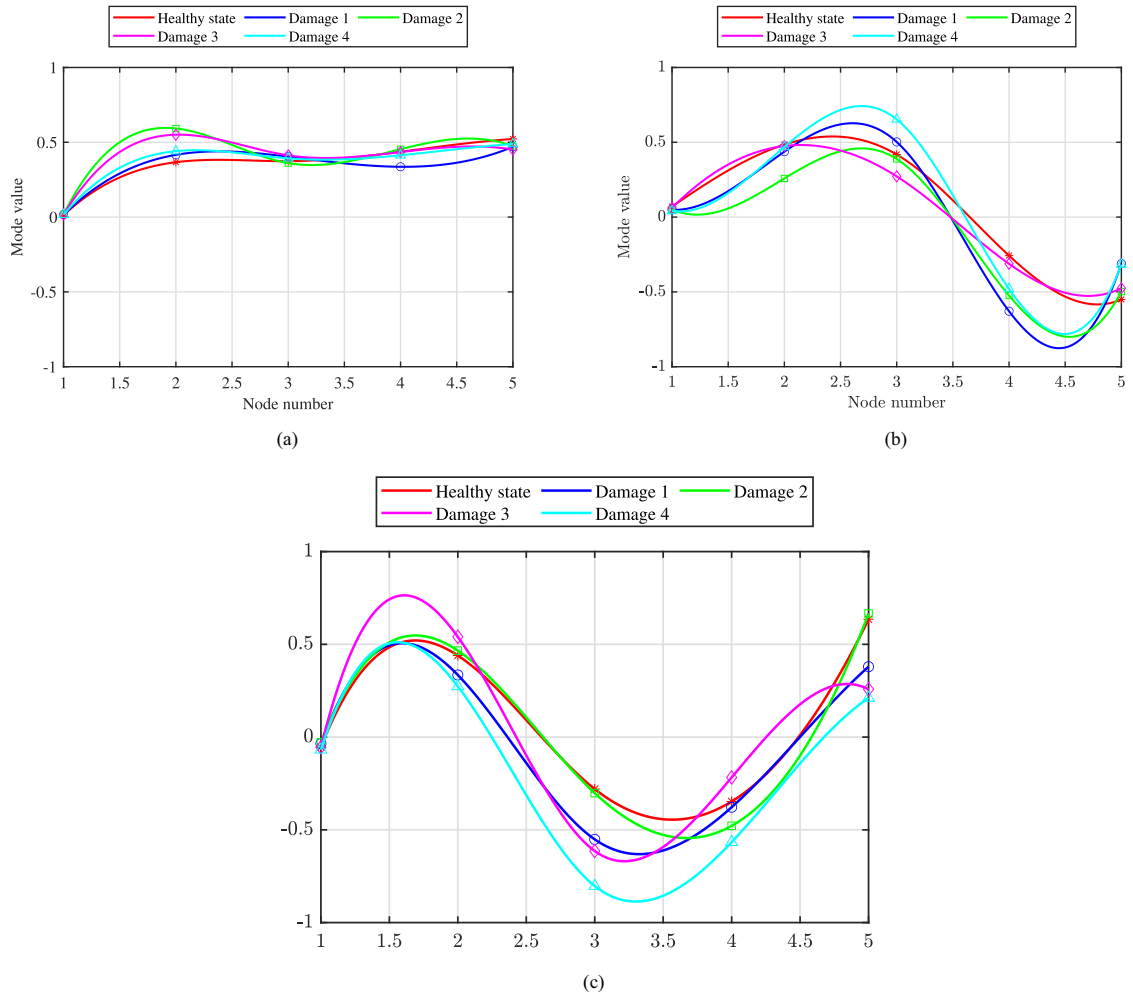


FIGURE 12. **Identified mode shapes of the suspension bridge model. (a) First mode. (b) Second mode. (c) Third mode.**

to identify the natural frequencies, while the corresponding first singular vectors were taken as estimates of the associated mode shapes. This approach is physically motivated by the fact that, near resonance, the vibration responses measured at different sensor locations become strongly correlated because they are governed by the same structural mode; therefore, the first singular value provides a clearer representation of the dominant global structural response and is more robust than relying on individual sensor PSDs.

Fig. 10 shows the values of the first three natural frequencies for the undamaged state. The first singular value highlights frequencies at which the structure vibrates in a coordinated manner, which usually correspond to natural frequencies.

The natural frequencies of the bridge for the four damage scenarios are shown in Fig. 11(a)–(d) respectively. The results reveal the following remarks:

1) *Damage 1:* The first natural frequency drops by 23.4%. This reflects localized stiffness degradation near the center of the bridge deck. The second and third frequencies, however, remain at 10.7 and 17.8 Hz, showing minimal variation (within $\pm$ 0.3%), implying no significant global damage.
2) *Damage 2:* The greatest drop is observed with the first mode falling by *47.6%*, and significant reductions in the second and third natural frequencies. This indicates a major structural disturbance and loss of symmetrical properties of the bridge stiffness.
3) *Damage 3:* The first frequency drops by *26.6%*, with moderate reductions in the second and third frequencies. The edge position of the damage causes global frequency shifts with moderate asymmetry.
4) *Damage 4:* A substantial drop in the first frequency of *46.1%* is recorded, showing severe global stiffness loss due to removal of one of the primary load-bearing cables.

The exact values of the natural frequencies at different states are listed in Table 5.

Using the equivalent stiffness-loss index defined in Section II, the first-mode frequency shifts were converted into approximate relative stiffness-reduction values, as summarized in Table 6.

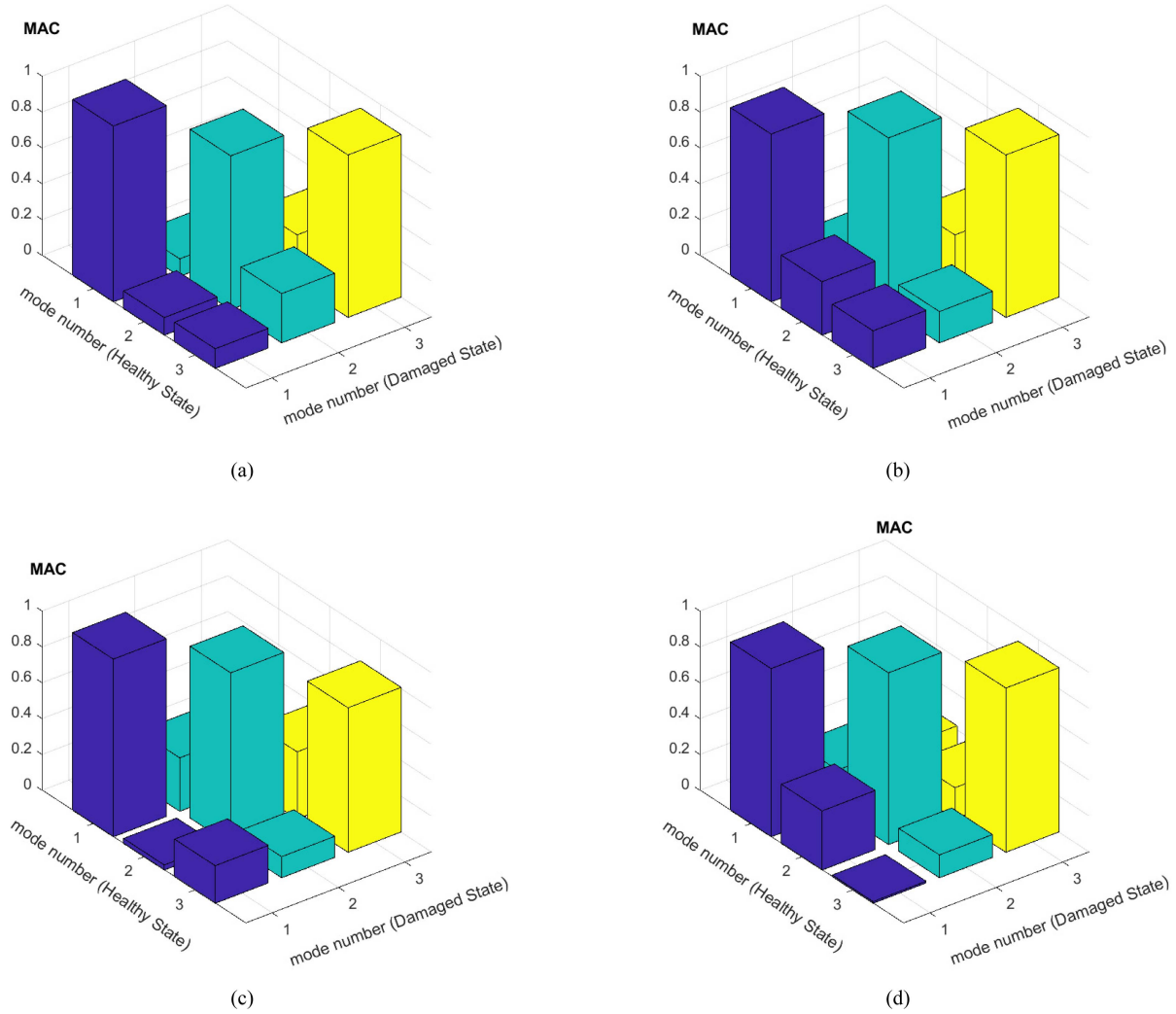


FIGURE 13. MAC matrices comparing the first three mode shapes of the healthy structure with those obtained under the four damage scenarios. Reduced diagonal values indicate increasing modal deviation caused by structural damage. (a) Damage scenario 1. (b) Damage scenario 2. (c) Damage scenario 3. (d) Damage scenario 4.

As shown in Table 6, the equivalent stiffness-loss index provides a clearer quantitative measure of damage severity than qualitative description alone. Damage 1 corresponds to a moderate stiffness reduction, while Damage 2 and Damage 4 produce the highest equivalent stiffness losses, indicating severe global degradation. Damage 3 shows an intermediate severity level. These values are consistent with the measured reductions in the first natural frequency and complement the mode-shape and MAC analyses by providing a simple scalar indicator of damage severity. These values should be interpreted as equivalent global stiffness-loss estimates derived from the first-mode frequency shift, and are, therefore, used as complementary severity indicators rather than direct localization measures.

C. MODE SHAPE ANALYSIS

The mode shape plots shown in Fig. 12 reveal the effect of structural damage on the dynamic behaviors of the system. These observed modal changes are consistent with the physical damage mechanisms described in Section IV-B: local hanger-removal cases produce more localized or asymmetric distortions, whereas main-cable damage leads to broader global stiffness effects.

1) *Damage 1* leads to subtle but noticeable changes near Sensors 2 and 3, where the hanger was removed, demonstrating that even minor suspension damage is detectable. Because only one hanger is removed, the loss of vertical support remains localized and does not substantially alter the global load path of the bridge, which explains the limited frequency change and the localized mode-shape deviation near the affected panel.
2) *Damage 2* caused the most significant distortions across all three modes. This includes peak amplitude shifts and curvature changes around the mid-span, indicative of high asymmetrical stiffness loss. These hangers directly connect the deck to the main cable and control local vertical stiffness. Removing two on opposite sides causes localized and asymmetric stiffness loss, leading to

uneven vertical deflections and torsional effects, which distort the mode shapes significantly.

3) *Damage 3* produces considerable distortions in the first and third modes, particularly near Node 1 and 2. This behavior is consistent with the removal of the edge main-cable segment, which weakens the effective boundary stiffness near the tower–cable interface and alters the local force-transfer path. As a result, the modes most sensitive to edge restraint and boundary conditions exhibit more pronounced shape distortion.
4) *Damage 4* shows global but scattered deviations in mode shapes, consistent with the redistribution of forces after removal of the main cable between towers. While they are key to maintaining the overall system in tension, the load is distributed, and the removal of one main cable can be partially compensated for by the other cable and the overall geometry of the bridge. The stiffness reduction is global and symmetric, which affects frequencies more than shapes.

D. MAC ANALYSIS

The MAC analysis is depicted in Fig. 13. The following remarks can be inferred from the analysis.

1) *Damage 1* shows consistently high MAC values across all modes, indicating minimal modal distortion. This validates that the effect of the damage is highly localized, with the overall structural dynamic behavior remaining largely intact.
2) *Damage 2* and *Damage 4* maintain strong diagonal MAC values, reflecting preservation of primary mode alignment. However, the presence of moderate off-diagonal elements suggests mode coupling effects and redistribution of stiffness due to either asymmetrical cable loss (Damage 2) or the removal of a key load-bearing element (Damage 4), introducing broader changes in structural response.
3) *Damage 3* leads to a more noticeable drop in the third mode's MAC value, indicating that higher order modes are particularly sensitive to disruptions at structural boundaries. This behavior reflects changes in the boundary condition at the tower-deck connection, which alters local stiffness and impacts mode shape coherence.

Overall, the MAC results are consistent with the underlying damage mechanisms: localized hanger-removal damage preserves much of the global modal correlation, whereas asymmetric or boundary-related cable damage introduces stronger modal deviation and coupling, and global main-cable damage affects the structure more broadly while still preserving partial mode alignment.

VI. CONCLUSION

In this work, an IoT-based SHM system for suspension bridges was developed and experimentally validated using synchronized vibration measurements acquired through GPS-based timing alignment and low-power sensing nodes. The proposed framework integrates distributed sensing, synchronized data acquisition, multitier wireless communication, cloud-based data processing, and vibration-based structural assessment within a unified architecture. The ESP-NOW/LTE communication design reduced communication energy consumption compared with TCP/IP over Wi-Fi and, together with the adopted sleep-mode strategy, contributed to extending node battery lifetime. Experimental validation on a laboratory-scale suspension bridge demonstrated reliable synchronized vibration acquisition and successful end-to-end operation of the proposed framework under different structural conditions. The collected measurements enabled the extraction of modal characteristics and captured damage-sensitive changes in the structural response, including natural-frequency shifts, mode-shape distortions, and MAC variations. These results confirm the feasibility of the proposed synchronized IoT-SHM framework for supporting continuous vibration monitoring and structural condition assessment under controlled laboratory conditions.

Overall, the proposed system provides a cost-effective, energy-efficient, and scalable platform for distributed structural monitoring. The experimental results successfully validated the integrated sensing, synchronization, communication, and cloud-processing framework, as well as its ability to capture damage-sensitive modal changes under controlled laboratory conditions using a scaled suspension bridge model. However, the long-term performance of the system under real environmental and operational conditions, as well as its deployment on full-scale suspension bridges, remains to be verified. A comprehensive repeatability assessment and long-term field validation therefore constitute important directions for future work.

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MUHAMMAD HASSAN (Student Member IEEE) received the B.Sc. degree in electrical engineering from Assiut University, Assiut, Egypt, in 2018.

He has been working as a Teaching and Research Assistant with the Electrical Engineering Department, Assiut University, since 2020. His current research interests include embedded systems design, the Internet of Things networks, cryptography, and structural health monitoring systems.



AMR A. NASSR received the B.Sc. degree in civil engineering from Assiut University, Assiut, Egypt, in 2004, and the M.A.Sc. and Ph.D. degrees in structural engineering from McMaster University, Hamilton, ON, Canada, in 2008 and 2012, respectively.

He is currently an Associate Professor with the Department of Civil Engineering, Ajman University, Ajman, UAE, and Assiut University. His research interests include structural health monitoring of structures and resiliency of critical infrastructures exposed to multiple hazards.

Dr. Nassr is a Member of many professional societies, including the Professional Engineers of Ontario (PEO), American Society of Civil Engineers (ASCE), and the Canadian Society of Civil Engineers (CSCE). He is also a former NSERC Fellow, a former JSPS Fellow, and the recipient of many awards, including the Queen Elizabeth II Award in Science and Technology and International Excellence Award.



MOHAMED ABDELRAHEEM (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees in electrical engineering from Assiut University, Assiut, Egypt, in 2004 and 2010, respectively, and the Ph.D. degree in electrical and computer engineering from Virginia Tech, Blacksburg, VA, USA, in 2015.

He is currently an Associate Professor of electrical engineering and computing science with the Rochester Institute of Technology (RIT) Dubai, UAE. He is also a Professor with the Electrical Engineering Department, Assiut University, from which he is currently on sabbatical leave. His research interests include wireless networking, Internet of Things (IoT), and embedded systems

Dr. AbdelRaheem is a Royal Academy of Engineering Leaders in Innovation Fellow. He has served as a Principal Investigator on several funded research projects supported by Science and Technology Development Fund (STDF) and Information Technology Industry Development Agency (ITIDA) in Egypt and by the Academic Research Committee (ARC) at Rochester Institute of Technology (RIT) Dubai.